Brady Shaw

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Education

Utah State University Logan, UT

B.S. in Statistics, Minors in Mathematics and Psychology *Expected Dec. 2026*

- Relevant Coursework: Data Technologies (SQL, Web Scraping, XML), Modern Regression Methods, Applied Machine Learning & Multivariate Statistics, Analysis of Designed Experiments, Mathematical Statistics, Probability, Linear Algebra, Introduction to R. *In progress:* Python for Data Science, Data Structures & Algorithms (Java), Statistical Visualization, Mathematical Methods for Data Science

Projects

Athlete Injury Risk Prediction | *Machine Learning, R, Statistical Modeling* Apr. 2026

- Built and cross-validated five classification models (KNN, LDA, QDA, Naïve Bayes, Logistic Regression) in R to predict next-season injury for 800 university football players from 18 physiological and performance features
- Tuned hyperparameters with 10-fold cross-validation and grid search on an 80/10/10 train/validation/test split, standardizing features with training-set parameters only to prevent data leakage
- Benchmarked against teammates' SVM, tree, random forest, and neural network models; QDA was the top model of all ten (97.5% validation, 96.25% held-out test accuracy), and logistic regression coefficients identified features linked to injury risk

Olympic Medal & IMDb Analysis | *R, Web Scraping, Regular Expressions* Fall 2025

- Scraped Wikipedia's all-time Olympic medal table (162 countries), repaired rows with merged ranks, and extracted country names and codes with regular expressions
- Scraped IMDb's Top 250 from a 2004 web archive and the live 2025 page, cleaned titles with regex, and joined the 113 films on both lists; 61% (69 of 113) rated higher in 2025

ParlGov Database Analysis | *SQL, R* Fall 2025

- Queried a relational database of elections and cabinets (37 countries, 1900–2023) in SQL from R, using joins, subqueries, and aggregation to answer multi-table questions

Live Population Growth & Simulation | *R, Web Scraping, Monte Carlo Simulation* Fall 2025

- Scraped a live world-population counter to estimate annual growth; the median of 10 runs was about 71.7M/yr, within 2% of the Census-reported 73M
- Ran a 10,000-run Monte Carlo simulation with a 99% confidence interval and validated it against derived results (mean 6.24 simulated vs. 6.25 exact)

Experience

Kitchen Design Specialist Feb. 2024 – Present

Home Depot *Logan, UT*

- Promoted twice since joining in Feb. 2024 (Millworks Associate → Flooring Specialist → Kitchen Design Specialist)
- Design customer kitchens in 20/20 design software and generate detailed quotes, contributing to $150K+ in annual revenue
- Analyze customer requirements and translate technical specifications into actionable recommendations

Full-Time Missionary Nov. 2021 – Nov. 2023

The Church of Jesus Christ of Latter-day Saints *Wisconsin*

- Coordinated logistics and travel arrangements for 150+ missionaries
- Led training programs and mentored new team members, strengthening leadership and communication skills

Technical Skills

Languages & Tools: R, SQL, Python (coursework in progress), XML, Web Scraping, Statistical Programming
Data Analysis: Regression Modeling, Monte Carlo Simulation, Hypothesis Testing, Exploratory Data Analysis, Machine Learning
Techniques: Data Cleaning, Regular Expressions, Data Visualization, Cross-Validation